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## SUMMARY (*cont*)

- The average acceleration is defined as $\vec{a}_{avg} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\Delta \vec{v}}{\Delta t}$ and instantaneous acceleration is defined as $\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$
- For constant acceleration, kinematic equations can be used to analyse motion of a particle.
- In projectile motion for which acceleration is uniform, the path traced by the particle is a parabola.
- The maximum height and range of the particle in projectile motion depend inversely on acceleration due to gravity g.
- The angular displacement of the particle is defined by $\theta = \frac{s}{r}$ and angular velocity $\vec{\omega} = \frac{d\theta}{dt}$
- The relation between the linear velocity and angular velocity is given by $\vec{v} = \vec{\omega} \times \vec{r}$
- The centripetal acceleration is given by $a_c = -\frac{v^2}{r}$ or $-\omega^2 r$ and is always directed towards the center of the circle.